AI DeskMate

Complete Feature Catalogue

An offline-first physical desk assistant built on an ESP32-S3, with a FastAPI sync hub, MongoDB Atlas persistence, Groq AI services, and a React dashboard that also ships as a Capacitor mobile app.

Project status

Core firmware, backend, dashboard, synchronization, voice pipeline, AI commands, weather, study notes, quizzes, and competitive productivity features are implemented in the repository. Battery integration, enclosure work, real ESP32-to-backend verification, and voice-based study quizzing remain incomplete or require additional hardware/testing.

Guiding principle

Local functions never depend on the network. Clock, timers, stopwatch, alarms, Pomodoro, touch, OLED, and relay control continue to work offline. AI and cloud services add conversation, complex commands, speech, and **learning features.**

Generated: 22 October 2026

1. Offline desk-assistant features

Digital clock

? Time, date, and day display
? NTP synchronization when Wi-Fi is available
? Continues working offline after synchronization
? Configurable timezone

Timer

? Start, pause, resume, and cancel
? Labels and remaining-time display
? Absolute deadline timing to avoid drift
? Completion notification/chime

Stopwatch

? Start and stop
? Lap/segment recording
? Reset and segment accumulation

Alarms

? Create, cancel, and trigger alarms
? Time-of-day and day-of-week repetition
? Local evaluation and persistent storage

Pomodoro / focus mode

? Focus and break phases; default 25/5-minute configuration
? Custom durations, pause, resume, stop, and skip
? Short-break and long-break cycles
? Session statistics and focus-time tracking

2. Local interaction and device controls

Reminders

? Text reminders with time-based triggering
? Create, update, and delete reminders

Task management

? Add, complete, and delete tasks
? Priority support
? Personal and competitive task types

OLED interface

? Home, Timer, Stopwatch, Alarms, Pomodoro, Tasks, and Reminders
? LED/relay, voice/listening, and network/status screens
? Wi-Fi, battery, focus-time, current-mode, and timer indicators

Touch controls

? Tap: change screen
? Double-tap: perform the current screen’s primary action
? Hold: push-to-talk voice interaction
? Automatic touch-polarity detection and hold-progress feedback

Relay and light control

? GPIO-controlled relay for an LED or other low-voltage load
? Toggle from the OLED, dashboard, or voice commands
? Persistent relay state and safe de-energized startup

Audio output

? Chimes and notification sounds
? MAX98357 amplifier and speaker support
? Non-blocking audio playback

3. Voice and AI features

Push-to-talk voice interaction

? Hold the touch sensor to speak
? INMP441 microphone capture
? Groq Whisper speech-to-text
? Edge TTS spoken responses

Natural-language command understanding

? Rule-based parsing for reliable common commands
? LLM fallback for flexible commands
? Examples: ?Set a timer for ten minutes?, ?Turn on the light?,
?It?s dark in here?, ?I need to focus?, and ?Wake me in 20 minutes?
? Conversational references such as ?remove the last task? and ?delete it?

Unified command pipeline

? Voice commands, dashboard chat, and button actions share one executor
? The same command behaves consistently from every surface

AI conversation

? General chat and open-ended questions
? Conversation memory using recent messages
? Short responses optimized for speech

Language support

? English, Hindi, and Hinglish commands
? Examples: ?Light chalu karo? and ?5 minute ka timer lagao?

AI fallback

? Mock provider for development
? Rule-based commands work without an API key
? Canned replies when Groq is unavailable

4. Weather, study, and learning

Weather

? Backend weather endpoint
? City or latitude/longitude configuration
? Cached responses to avoid excessive upstream requests
? Stale-cache fallback when the weather provider is unavailable
? Firmware weather-service support

Study Notes

? Upload PDF, Word, Excel, and plain-text notes
? Extract note text on the backend
? Store notes by device/user
? Generate questions grounded only in uploaded notes
? Submit answers and receive grades
? Track quiz sessions, accuracy, and perfect quizzes

Current limitation

The dashboard study workflow is implemented. Voice commands such as ?quiz me on my biology notes? are planned as a follow-up. OCR for scanned **images is outside the current scope.**

5. Competitive productivity

Experience and progression

? XP from Pomodoro focus sessions, completed tasks, and quiz answers
? Bonus XP for perfect quizzes
? User levels and activity streaks

Leagues

? Bronze, Silver, Gold, Platinum, and Diamond
? League determined from genuine activity-derived XP

Statistics

? Focus time and completed focus sessions
? Personal and competitive task totals
? Quiz questions, correct answers, accuracy, and perfect quizzes
? Activity history and daily streak tracking

Friend competition

? Competitor/referral codes
? Friend challenges
? Incoming and outgoing invitations
? Accept, decline, and cancel challenge requests
? Competitive task lists
? Leaderboard

Implemented in the dashboard Competitive page and backend competitive API.

6. Dashboard and mobile-app features

Dashboard pages

? Overview; AI Chat; Timer; Stopwatch; Alarms; Pomodoro
? Tasks; Reminders; Study Notes; Analytics
? Competitive productivity; Device Status; Device Settings

Dashboard controls

? Start, pause, resume, and cancel timers
? Manage tasks, reminders, and alarms
? Control Pomodoro sessions and the relay/light
? View device telemetry, OLED state, voice commands, and analytics
? Configure the backend API address and device settings

Web and mobile

? React + Vite + TypeScript + Tailwind web application
? Android app through Capacitor
? iOS-ready Capacitor project
? Responsive desktop and mobile layouts
? Mobile bottom navigation and More sheet
? Native status bar, haptics, resume handling, and back-button support
? Hash routing for native builds and LAN development support

7. Real-time synchronization

Backend sync hub

? FastAPI REST API and WebSocket communication
? ESP32-to-dashboard event propagation
? Dashboard-to-device commands
? Device heartbeat, telemetry, and online/offline status

Reliable state reconciliation

? Full snapshot fetched on load and WebSocket reconnect
? No missed-event replay complexity
? Device and dashboard share one authoritative state model
? Server-time correction for client clock skew

Connection resilience

? Exponential reconnect backoff with jitter
? Immediate retry on focus or network recovery
? Ping/pong watchdog for stale mobile connections

Timer synchronization

? Timers use absolute ends_at timestamps
? Clients derive countdowns locally
? No per-second timer polling over the network
? Countdown remains smooth during poor connectivity
? Correct state is recovered after a suspended tab or app reconnect

8. Backend and platform capabilities

Backend

? FastAPI with asynchronous PyMongo and MongoDB Atlas
? CRUD APIs for devices, timers, stopwatches, alarms, reminders, tasks, Pomodoro, notes, quizzes, settings, chat, analytics, and competition
? Background scheduler for timers, reminders, alarms, and Pomodoro phases
? Automatic indexes and TTL cleanup for device events
? OpenAPI/Swagger documentation and health reporting
? Explicit degraded/503 behavior when the database is unavailable

AI services

? GroqCloud LLM integration
? Whisper speech recognition
? Edge TTS speech generation
? Environment-based provider and model configuration

Security options

? Optional API-key authentication
? Optional device-token authentication
? WebSocket token authentication
? Environment-based secret handling
? Note: VITE_ variables are compiled into the client and are not secrets

9. Accessibility, hardware, and status

Accessibility and UX

? Keyboard navigation and visible focus indicators
? Skip link and labelled form controls
? Accessible switch controls
? Screen-reader table alternative for charts
? Polite toast live region
? Reduced-motion support
? Full WCAG conformance still needs manual assistive-technology testing

Hardware platform

? ESP32-S3 controller with Wi-Fi
? SH1106 OLED over I2C
? INMP441 digital microphone
? MAX98357 amplifier and speaker
? TTP223 capacitive touch sensor
? GPIO-controlled relay and LED/load
? Persistent NVS storage
? Planned 18650 battery system with TP4056 and MT3608

Known incomplete or unverified items

? Battery rail still needs MT3608 output trim to 5.0 V and final wiring
? Enclosure/final physical integration has not been completed
? Real ESP32-to-backend connection has not been fully verified
? Android app was generated/synced but not installed on a handset here
? Voice-based study-note quizzing remains a planned follow-up

Reference files

README.md, backend/README.md, dashboard/README.md, docs/AI-DeskMate-Project-Reference.md, and docs/FIRMWARE.md